Introduction and Math Review Practice 3

Module 1 - Introduction & Math Review · Daily practice sheet · No date

Name _____ Date _____

1a 1. The table below shows the decomposition of a drug as a function of time.

Time (minute)	Drug A (mg)
10	97.0
20	89.0
40	73.0
60	57.0
90	34.0
120	10.0
130	2.5

a. How would you classify the decrease in the amount of drug A?

A. First-order
B. Neither order fits the data
C. Zero-order

1b 1. The table below shows the decomposition of a drug as a function of time.

Time (minute)	Drug A (mg)
10	97.0
20	89.0
40	73.0
60	57.0
90	34.0
120	10.0
130	2.5

b. What is the rate constant for the decrease in drug?

1c 1. The table below shows the decomposition of a drug as a function of time.

Time (minute)	Drug A (mg)
10	97.0
20	89.0
40	73.0
60	57.0
90	34.0
120	10.0
130	2.5

c. What is the half-life of drug decomposition?

2a 2. A solution of a drug was freshly prepared at a concentration of 500 mg/mL. After 60 days at 25°C, the drug concentration in the solution was 150 mg/mL. a. Assuming first-order kinetics, when will the drug decline to one-half of the original concentration?

2b 2. A solution of a drug was freshly prepared at a concentration of 500 mg/mL. After 60 days at 25°C, the drug concentration in the solution was 150 mg/mL. b. Assuming zero-order kinetics, when will the drug decline to one-half of the original concentration?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

1a Her key

Answer: C. Zero-order

WHAT THE STEM GIVES

A at 20 min	89.0	mg; the start of a 20-minute interval
A at 40 min	73.0	mg; the end of one interval and the start of the next
A at 60 min	57.0	mg; the end of the second interval
A at 10, 90, 120, 130 min	97.0, 34.0, 10.0, 2.5	mg; not needed: the two 20-minute intervals decide the order

THE WORKING, ONE LINE AT A TIME

SETUP

Equal intervals: 20 to 40 min and 40 to 60 min, 20 minutes each

The order is read from how the amount falls over equal time intervals: equal differences mean zero order, equal ratios mean first order.

ALGEBRA

$89.0 - 73.0 = 16.0$ mg; $73.0 - 57.0 = 16.0$ mg

The same 16.0 mg is lost in each 20-minute interval: a constant amount per unit time, the zero-order pattern.

ALGEBRA

$$\frac{89.0}{73.0} = 1.219; \quad \frac{73.0}{57.0} = 1.281$$

The two ratios differ, so the fraction lost per interval is not constant and the decline is not first order.

> **DOES THE NUMBER MAKE SENSE?**
>
> A zero-order amount falls along a straight line: 16.0 mg every 20 minutes here. The ratios 1.219 and 1.281 differ, so ln(amount) against time would curve, which rules out first order.

The drug is decreasing at a constant amount per unit time: 16.0 mg in each 20 minutes from 20 to 60 minutes. Equal amounts lost in equal times is zero order.

1b Her key

Answer: 0.8 mg/min

WHAT THE STEM GIVES

A at 20 min	89.0	mg; the first point of her slope

A at 40 min	73.0	mg; on the same line: 16.0 mg lost in each 20 minutes
A at 60 min	57.0	mg; the second point: 32 mg lost in 40 minutes
A at 10, 90, 120, 130 min	97.0, 34.0, 10.0, 2.5	mg; any two rows give 0.75 to 0.8 mg/min

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = -\text{slope of the line} = -\frac{\Delta Y}{\Delta X}$$

Zero order: A = A0 − kt is a straight line in the amount, so the rate constant is the slope with the sign reversed.

ALGEBRA

$$k = -\frac{(57 - 89)\ \text{mg}}{(60 - 20)\ \text{min}} = \frac{32\ \text{mg}}{40\ \text{min}} = 0.8\ \text{mg/min}$$

Her key takes the 20 and 60 minute points. mg over min gives mg/min: a fixed 0.8 mg of drug A is lost each minute, an amount per time because the order is zero.

> **DOES THE NUMBER MAKE SENSE?**
> From 20 to 60 minutes the amount falls from 89.0 to 57.0 mg, a little under 1 mg each minute; 0.8 mg/min fits, and at that rate 40 minutes removes 32 mg.

Any two rows of the table give a slope between 0.75 and 0.8 mg/min, and any of them is accepted; her key uses the 20- and 60-minute rows.

1c Her key

Answer: 65.6 min

WHAT THE STEM GIVES

k	0.8 mg/min	adds back 0.8 × 40 = 32 mg, then sits in the 2k denominator
A at 40 min	73.0	mg; the measured amount from which A0 is recovered
A at 10, 20, 60, 90, 120, 130 min	97.0, 89.0, 57.0, 34.0, 10.0, 2.5	mg; any one row gives A0 between 105 and 106.5 mg

THE WORKING, ONE LINE AT A TIME

SETUP

$$A0 = A + kt$$

A = A0 − kt rearranged: the amount at time zero is a later amount plus what was lost in between, k × t, because a zero-order process loses a fixed amount each minute.

ALGEBRA

$$A0 = 73\ \text{mg} + (0.8\ \text{mg/min})(40\ \text{min}) = 105\ \text{mg}$$

Her key uses the 40-minute point. The minutes cancel in (mg/min)(min), so 32 mg is added to 73 mg: drug A started at 105 mg.

SETUP

$$t½ = \frac{A0}{2k}$$

Half of A0 must be lost at k mg each minute, so the time is $\frac{A0}{2}$ divided by k. A zero-order half-life depends on the starting amount.

ALGEBRA

$$t½ = \frac{105\ \text{mg}}{2(0.8\ \text{mg/min})} = 65.6\ \text{min}$$

mg over mg/min leaves minutes: $\frac{105}{1.6}$ = 65.625 min, reported as 65.6 min.

> **DOES THE NUMBER MAKE SENSE?**
> Drug A started at 105 mg; at 60 minutes the table still shows 57.0 mg, just over half of 105 mg, so the half-life is a little over 60 minutes, and 65.6 min fits.

2a Her key

Answer: 34.5 days

WHAT THE STEM GIVES

C_0	500 mg/mL	the starting concentration, top of the ratio
C	150 mg/mL	after 60 days, bottom of the ratio
t	60 days	the denominator of k
temperature	25°C	*not needed: held constant*

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = -\frac{\Delta Y}{\Delta X} = -\frac{\ln(150) - \ln(500)}{60\ \text{days}}$$

First order: ln C = ln C_0 − kt is a straight line, so k is minus the slope of ln C against time, from the two concentrations 60 days apart.

ALGEBRA

$$k = \frac{\ln(500 \div 150)}{60 \text{ days}} = \frac{1.204}{60 \text{ days}} = 0.02007 \text{ day}^{-1}$$

The log of a ratio of two concentrations in mg/mL is a pure number, 1.204, so dividing by days leaves day^{-1}. Her key rounds k to 0.02 day^{-1}.

ALGEBRA

$$t½ = \frac{0.693}{0.02007 \text{ day}^{-1}} = 34.5 \text{ days}$$

$t½ = \frac{0.693}{k}$: a pure number divided by day^{-1} gives days. Carried unrounded, k gives 34.54 days, her 34.5.

DOES THE NUMBER MAKE SENSE?

After 60 days 150 of the 500 mg/mL remains, less than half but more than a quarter, so 60 days is between one and two half-lives and t½ lies between half of 60 days and 60 days; 34.5 days fits.

Her key prints k = 0.02 day^{-1} and t½ = 34.5 days. 0.693 divided by the rounded 0.02 is 34.65 days; her 34.5 follows from the unrounded k, 0.02007 day^{-1}. Both are accepted.

2b Her key

Answer: 42.9 days

WHAT THE STEM GIVES

C_0	500 mg/mL	the starting concentration, the numerator of t½
C	150 mg/mL	after 60 days: 350 mg/mL lost
t	60 days	the denominator of k
temperature	25°C	*not needed: held constant*

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = -\frac{\Delta Y}{\Delta X} = -\frac{(150 - 500) \text{ mg/mL}}{60 \text{ days}}$$

Zero order: $C = C_0 - kt$ is a straight line in the concentration itself, so k is minus the slope of C against time.

ALGEBRA

$$k = \frac{350 \text{ mg/mL}}{60 \text{ days}} = 5.83 \text{ mg/mL/day}$$

mg/mL over days leaves mg/mL/day: a fixed concentration lost each day, the zero-order unit.

ALGEBRA

$$t½ = \frac{C_0}{2k} = \frac{500 \text{ mg/mL}}{2(5.83 \text{ mg/mL/day})} = 42.9 \text{ days}$$

The mg/mL cancel and days are left. The zero-order half-life uses the starting concentration, 500 mg/mL, because half of it must be lost at a fixed 5.83 mg/mL each day.

DOES THE NUMBER MAKE SENSE?

> Losing 5.83 mg/mL each day, the drug loses half of 500 mg/mL in a little over six weeks, so 42.9 days fits; it must be under the 60 days after which 150 mg/mL, less than half, remained.
